arr = {1, 6, 11, 5}

n = 4

// Step 1: Calculate total sum

sum = 1 + 6 + 11 + 5 = 23

// Step 2: Initialize DP

dp[0...23] = false

dp[0] = true

// Initial DP:

// {T, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F}

// Step 3: Process elements

// i = 0 → element = 1

// Update dp[j] from j = 23 → 1

dp[1] = dp[0] → true

// DP now:

// {T, T, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F}

// i = 1 → element = 6

dp[6] = dp[0] → true

dp[7] = dp[1] → true

// DP now:

// {T, T, F, F, F, F, T, T, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F}

// i = 2 → element = 11

dp[11] = dp[0] → true

dp[12] = dp[1] → true

dp[17] = dp[6] → true

dp[18] = dp[7] → true

// DP now:

// {T, T, F, F, F, F, T, T, F, F, F, T, T, F, F, F, F, T, T, F, F, F, F, F}

// i = 3 → element = 5

dp[5] = dp[0] → true

dp[6] = dp[1] → true (already true)

dp[7] = dp[2] → false (no change)

dp[11] = dp[6] → true

dp[12] = dp[7] → true

dp[16] = dp[11] → true

dp[17] = dp[12] → true

dp[22] = dp[17] → true

dp[23] = dp[18] → true

// Final DP:

// {T, T, F, F, F, T, T, T, F, F, F, T, T, F, F, F, T, T, T, F, F, F, T, T}

// Step 4: Find closest to sum/2 = 11

// Possible j where dp[j] = true and j ≤ 11:

// 0, 1, 5, 6, 7, 11

j = 11 (closest)

// Step 5: Compute answer

min_diff = 23 - 2*11 = 1